

ECE 444 Equation Sheet - Fall 2026 (Modules 1-3)

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Wave Fundamentals and the Far Field

$$k = \frac{2\pi}{\lambda}, \quad c = 3 \times 10^8 \text{ m/s}$$

$$\eta_0 = \frac{|\mathbf{E}|}{|\mathbf{H}|} \approx 377 \, \Omega \text{ (free space)}$$

$$\text{outgoing wave: } \frac{e^{-jkr}}{r}; \text{ field } \propto \frac{1}{r}, \text{ power } \propto \frac{1}{r^2}$$

$$P_{\text{avg}} = \frac{1}{2} \text{Re} \left\{ \mathbf{E} \times \mathbf{H}^* \right\}; \quad S_{\text{rad}} = \frac{|\mathbf{E}|^2}{2\eta_0}$$

$$U(\theta, \phi) = r^2 S_{\text{rad}} \text{ (W/sr); whole sphere} = 4\pi \text{ sr}$$

Pattern, Directivity, Gain

$$|F(\theta, \phi)| \text{ normalized field pattern; isotropic: } D=1, \text{ 0 dBi}$$

$$D(\theta, \phi) = \frac{4\pi U}{P_{\text{rad}}}; \text{ pencil beam } \approx \frac{41,253}{\theta_1^\circ \theta_2^\circ} \text{ (real 26k--32k)}$$

$$G = \eta_{\text{rad}} D, \quad \eta_{\text{rad}} = \frac{P_{\text{rad}}}{P_{\text{in}}}; \quad \text{dBi} = \text{dBd} + 2.15$$

$$G_{\text{re}} = \left(1 - |\Gamma|^2\right) G; \text{ VSWR 2:1 costs 11\% } (\approx 0.5 \text{ dB})$$

$$\theta_{\text{HP}} \text{ (HPBW) at } -3 \text{ dB}; \quad \text{FNBW} \approx 2 \theta_{\text{HP}}$$

$$A_e = \frac{\lambda^2 G}{4\pi} = \eta_{\text{ap}} A_{\text{phys}}$$

$$\text{reciprocity: } A_e/G = \lambda^2/4\pi \text{ for every antenna}$$

Polarization

$$\text{fixed by } E_x, E_y, \delta; \quad \text{AR} = \frac{\text{major axis}}{\text{minor axis}}$$

$$0 \text{ dB} = \text{pure CP}; \quad \infty = \text{pure linear}; \quad \text{hardware} \leq 3 \text{ dB}$$

$$\text{PLF} = \left| \hat{\rho}_{\text{w}} \cdot \hat{\rho}_{\text{a}}^* \right|^2; \quad \cos^2 \psi \text{ for two linears}$$

$$\text{CP vs. linear, AR=0 dB : flat } -3 \text{ dB}; \text{ AR=3 dB : } -1.8 \text{ to } -4.8 \text{ dB}$$

$$\text{wrong circular sense: AR=3 dB} \rightarrow -9.6 \text{ dB}, \text{ AR=6 dB} \rightarrow -4.5 \text{ dB}$$

$$\text{FBW} = \frac{f_H - f_L}{f_c}; \quad \text{Chu-Harrington: FBW} \approx 1/Q \text{ (} k a=1, Q \approx 2 \text{)}$$

Impedance, VSWR, Matching

$$Z_{\text{in}} = R_{\text{in}} + jX_{\text{in}} \text{ (resonance: } X_{\text{in}}=0\text{); } \eta_{\text{rad}} = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}}$$

$$\Gamma = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0}; \quad \text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

$$\text{VSWR} \leq 2 \leftrightarrow \text{RL} \geq 9.5 \text{ dB} \leftrightarrow 11\% \text{ reflected}$$

$$\text{mismatch loss} = -10 \log_{10} \left(1 - |\Gamma|^2\right) \leq 0.5 \text{ dB at VSWR}=2$$

$$Z_{\text{in}} = Z_0 \frac{Z_L + Z_0 \tanh(j\gamma\ell)}{Z_0 + Z_L \tanh(j\gamma\ell)}; \text{ lossless: tan for tanh}$$

$$\Gamma(z) = \Gamma_L e^{2\gamma z}; \quad \beta = \omega \sqrt{\mu\epsilon} = 2\pi/\lambda$$

$$\text{quarter-wave transf.: } Z_1 = \sqrt{Z_0 R_L}$$

$$\text{L-match } Q = \sqrt{R_{\text{big}}/R_{\text{small}} - 1}; \text{ balun 4:1: } 300 \, \Omega \rightarrow 75 \, \Omega$$

Field Regions

$$\text{one dipole eqn, three terms: } 1 : \frac{1}{kr} : \frac{1}{(kr)^2}; \text{ } kr=1 \text{ crossover}$$

$$\text{true far field: } kr \gg 1$$

$$\text{reactive near-field edge: } r = 0.62 \sqrt{D^3/\lambda}$$

$$\text{far-field distance: } r \geq \frac{2D^2}{\lambda} \text{ (} D=\text{largest dim)}$$

$$\text{phase-error budget: } \Delta \leq \lambda/16, \pi/8 \text{ (} 22.5^\circ \text{)}$$

Radiation Integrals

$$\mathbf{A} \text{ (vector potential), } \mathbf{B} = \nabla \times \mathbf{A}; \text{ one integral, two curls}$$

$$\mathbf{N}(\theta, \phi) = \int \mathbf{J} e^{+jk\hat{\mathbf{r}} \cdot \mathbf{r}'} dV'$$

$$E_\theta = -j\omega\mu \frac{e^{-jkr}}{4\pi r} N_\theta, \quad E_r \approx 0$$

$$\text{far-field approx: keep } \hat{\mathbf{r}} \cdot \mathbf{r}' \text{ in phase, drop in ampl.}$$

$$U(\theta, \phi) = \frac{\eta_0 k^2}{32\pi^2} \left(|N_\theta|^2 + |N_\phi|^2 \right); \quad I(z') \leftrightarrow N_z(k_z) \text{ Fourier pair}$$

Wire Antennas

$$\text{infinitesimal dipole: } |F| = \sin \theta; \text{ HPBW } 90^\circ, D=1.5 \text{ (1.76 dBi)}$$

$$\text{general dipole: } |F| \propto \left| \frac{\cos\left(\frac{kL}{2} \cos \theta\right) - \cos \frac{kL}{2}}{\sin \theta} \right|$$

$$\text{half-wave dipole: } |F| = \frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta}; \text{ HPBW } 78.1^\circ, D=1.64 \text{ (2.15 dBi)}$$

$$Z_{\text{in}} \approx 73 + j42.5 \, \Omega; \text{ resonant } \approx 0.47\text{--}0.48\lambda, \approx 70 \, \Omega \text{ real}$$

$$5\% \text{ rule: } \frac{143}{f_{\text{MHz}}} \text{ m; longer dipoles: peak } \approx 5.2 \text{ dBi near } 1.25\lambda$$

$$R_{\text{rad}} \text{ (short dipole) : } 20\pi^2 (\ell/\lambda)^2$$

$$\text{quarter-wave monopole: } Z_{\text{in}} = \frac{1}{2} Z_{\text{in}}^{\text{dipole}} \approx 36.5 + j21.3 \, \Omega, D=5.15 \text{ dBi}$$

$$\text{small loop: } |F| = \sin \theta, D=1.5 \text{ (1.76 dBi)}; R_r = 20\pi^2 (C/\lambda)^4$$

$$\text{resonant loop, } C \approx 1\lambda : 100\text{--}130 \, \Omega, \approx 3.1 \text{ dBi}$$

$$\text{Chu limit: } Q \gtrsim 1/(ka)^3$$

Apertures and Patches

$$G = \eta_{\text{ap}} \frac{4\pi A}{\lambda^2}; \quad \eta_{\text{ap}} \approx 0.5\text{--}0.7 \text{ (horn } \approx 0.5, \text{ reflector } 0.55\text{--}0.7)$$

$$\text{circular dish: } G = \eta_{\text{ap}} (\pi D/\lambda)^2, +6 \text{ dB /doubling; } \theta_{\text{HP}} \approx 70^\circ \lambda/D$$

$$f/D : 0.3\text{--}0.6 \text{ typ.}; \text{ edge taper } \approx -10 \text{ dB}$$

$$\text{Ruze: } 685.8(\sigma/\lambda)^2 \text{ dB}; \lambda/50 \rightarrow 0.27 \text{ dB}, \lambda/16 \rightarrow 2.7 \text{ dB}$$

$$\text{patch: } L \approx \lambda_d/2, \text{ shorten } 2\Delta L; \text{ broadside } 5\text{--}8 \text{ dBi}, \text{ BW } 1\text{--}5\%$$

$$\text{slot: } Z_{\text{slot}} Z_{\text{dipole}} = \eta_0^2/4 \text{ (Babinet); resonant } \approx 485 \, \Omega$$

$$\text{horn: optimum } \lambda/4 \text{ E-plane, } 3\lambda/8 \text{ H-plane}$$

$$\text{Yagi: } +3 \text{ dB /boom doubling, } 8\text{--}15 \text{ dBi}$$

$$\text{array gain over 1 elem.: } 10 \log_{10} N$$

Link Budget

$$\text{EIRP} = P_t G_t; \quad \text{FSPL} = 20 \log_{10} \left(\frac{4\pi R}{\lambda} \right)$$
$$\text{Friis: } P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2; \quad 2.4 \text{ GHz ex.: } 156 \text{ dB loss}, -90 \text{ dBm}$$

radar range eqn.: deferred to Module 4/L29 (preview only)

Aperture Distributions and Tapers

$E_a(x)$ aperture distribution; $u = \frac{L}{\lambda} \sin \theta$; pattern = $\mathcal{F}\{E_a(x)\}$

uniform: $|F| = \left| \frac{\sin \pi u}{\pi u} \right|$; $\theta_{\text{HP}} \approx 0.886\lambda/L$

1st sidelobe uniform: $-13.3 \text{ dB (line)}, -17.6 \text{ dB (circ.)}$

taper $\eta_{\text{ap}}/\text{SLL}/\theta_{\text{HP}} \times$: uniform 1.00 / -13.3 dB / 1.00

cosine 0.81 / -23 dB / 1.19; triangular 0.75 / -26.5 dB / 1.27

cosine² 0.667 / -31.5 dB / 1.44

taper trade: $-13.3 \rightarrow -31.5 \text{ dB}$ costs 63% beamwidth, 1.8 dB

8-elem. presets, HPBW/peak drop/ η_t : Uniform 13.1° / 0 dB / 1.00

Hann 19.5° / -4.7 dB / 0.75 (−1.2 dB)

Blackman 23.1° / -6.1 dB / 0.66 (−1.8 dB)

Chebyshev 24.3° / -6.5 dB / 0.62 (−2.1 dB)

$\eta_t = \frac{(\sum a_n)^2}{N \sum a_n^2}$; peak drop = $20 \log_{10} \left(\sum a_n / N \right) \neq \eta_t$

Chebyshev (Dolph), -30 dB on $N=8:26, 52, 81, 100 \%$

HPBW mult. vs. uniform:

1.12 (−20 dB), 1.30 (−30 dB), 1.42 (−40 dB)

Array Factor, Steering, Beamwidth

θ, θ_0 from broadside; $AF(\theta) = \sum_n a_n e^{j n k d \sin \theta}$

$\psi = kd(\sin \theta - \sin \theta_0)$; $AF_N(\psi) = \frac{\sin(N\psi/2)}{N \sin(\psi/2)}$ (peak=1)

nulls at $\psi=2\pi m/N$; 1st sidelobe unif.: -13 dB (−12.8 dB at $N=8$)

$F(\theta) = EF(\theta) AF(\theta)$ (pattern mult.)

$\theta_{\text{HP}} \approx \frac{0.886 \lambda}{Nd \cos \theta_0}$; FNBW = $2 \arcsin \frac{\lambda}{Nd}$; PHASER: 13.2°/30.1°

$D \approx \frac{2Nd}{\lambda}$ (broadside ULA; 8.9 dBi at $N=8$)

$\Delta\phi = kd \sin \theta_0$ (elem. phase); $\phi_n = -n \Delta\phi$

$\sin \theta_{\text{null}} = \sin \theta_0 \pm \frac{m\lambda}{Nd}$

$\theta_{\text{HP}}(\theta_0) \approx \frac{\theta_{\text{HP}}(0)}{\cos \theta_0}$; scan loss $\propto \cos \theta_0$: −0.6, −1.5, −3.0 dB at 30°, 45°, 60°

$G_e(\theta) = G_e(0) \cos \theta$; $G_{\text{total(dB)}} = G_{\text{element}} + D_{\text{array}}$

$d < \frac{\lambda}{1 + |\sin \theta_0|}$ for scan volume; $\lambda/2$ scans anywhere

Grating Lobes, Squint, Quantization

$\sin \theta_g = \sin \theta_0 \pm \frac{m\lambda}{d}$; avoid: $d < \frac{\lambda}{1 + |\sin \theta_0|}$, $d \leq \lambda/2$ safe

grating lobe height = main lobe (taper doesn't reduce it)

$\Delta\theta = \arcsin \left(\frac{f_0}{f} \sin \theta_0 \right) - \theta_0$; 500 MHz off at 45° $\rightarrow +2.9^\circ$

$\Delta\theta \approx -\frac{\Delta f}{f_0} \tan \theta_0$; true time delay removes squint

LSB = $\frac{360^\circ}{2^B}$; ADAR1000 $B=7 \rightarrow \text{LSB} = 2.8125^\circ$

$\Delta\theta_0 \approx \frac{\text{LSB}}{360^\circ d/\lambda} \approx 0.91^\circ$ on the PHASER

QSSL $\approx -6B \text{ dB}$; 2 bits $\rightarrow -12 \text{ dB}$, 7 bits $\rightarrow -42 \text{ dB}$

Null Steering

$v_n(\theta) = e^{j n k d \sin \theta}$; $\mathbf{w}_d = \mathbf{v}^*(\theta_0)$

$\mathbf{w} = \mathbf{w}_d - r_n \mathbf{w}_n$; $r_n = \frac{\mathbf{w}_n^H \mathbf{w}_d}{\mathbf{w}_n^H \mathbf{w}_n}$

$|r_n| = 0.225$ at -13 dB SLL

null + 22.5°: cost 2.0 dB theory, 1.8 dB meas.

max. simultaneous nulls: $N - 1$

MVDR: $\mathbf{w} = \frac{R^{-1} \mathbf{s}}{\mathbf{s}^H R^{-1} \mathbf{s}}$; 17–19 dB suppression

$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$, $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$

$c = 3 \times 10^8 \text{ m/s}$